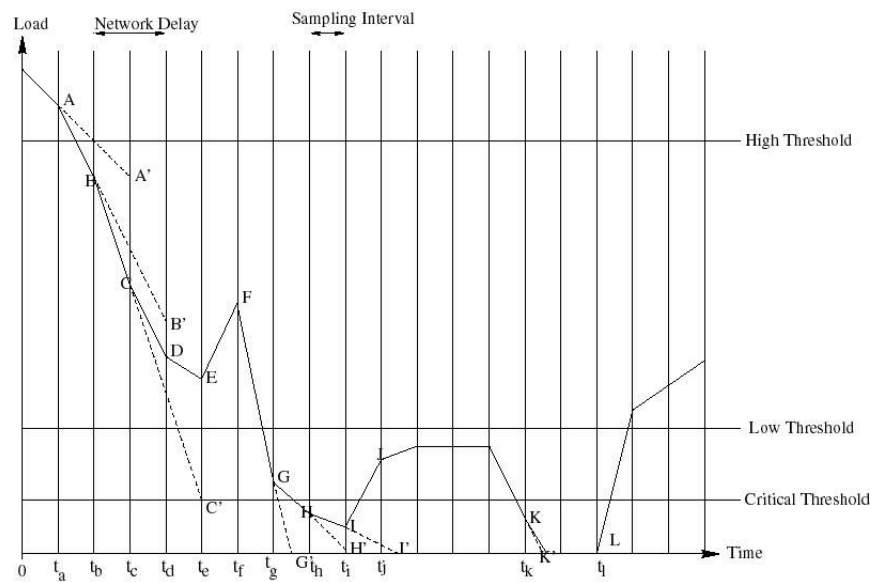


RoC-LB depicted in time



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RoC-LB: Where Phase

- ▶ Sink processor initiates process
 - ▶ Select source processor from table
 - ▶ Sends request to selected processor
 - ▶ Deletes source entry from table
 - ▶ Upon receiving reply updates table



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RoC-LB: Where Phase

- ▶ Receiving processor
 - ▶ Transfers tasks to sink if still source
 - ▶ Increments count
 - ▶ Forward request to another source processor if no longer source or not able to satisfy whole request
 - ▶ Updates table based on request message

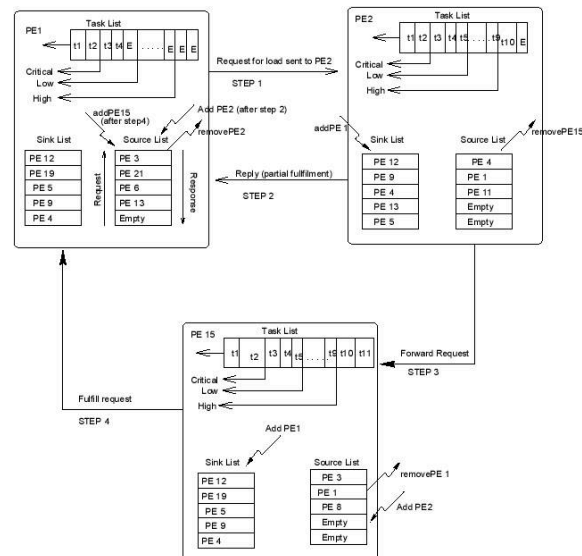


RoC-LB: Transfer request protocol

- ▶ Request Fields:
 - ▶ Sink processor identification
 - ▶ number of load units
 - ▶ load state of preceding processor
 - ▶ count number



ROC-LB: Transfer request protocol



RoC-LB: How many / Which Phases

- ▶ Sender Side: Assuming DL constant for the duration of a ND calculate Predicted Load (PL). Request $\text{abs}(\text{PL})$
- ▶ Receiver Side: Transfer tasks above HT
- ▶ Older Tasks. Higher probability of living long enough to amortize their migration cost



ROC-LB: Classification

- ▶ Dynamic
- ▶ Distributed
- ▶ On demand
- ▶ Preemptive
- ▶ Implicit



RoC-LB: Experiments

- ▶ Each simulation uses 32 000 tasks.
- ▶ Tasks have different sizes and life span.
- ▶ Four different Load Balancing Algorithms used.
- ▶ Four different network topologies:
 - ▶ Mesh
 - ▶ Hypercube
 - ▶ Fully connected network
 - ▶ Network of Workstations



RoC-LB: Performance Metrics

Normalized Performance

$$NP = \frac{T_{noLB} - T_{LB}}{T_{noLB} - T_{opt}}$$



RoC-LB: Results

	Symmetric distribution			
	Mesh	HyperCube	Fully Connected	Network of Workstations
Gradient Model	0.66	0.69	0.73	0.52
Sender Initiated Diffusion	0.70	0.71	0.74	0.59
Central Job Dispatcher	0.59	0.62	0.73	0.37
Rate of Change Model	0.79	0.79	0.87	0.61



RoC-LB: Results

Asymmetric distribution

	Mesh	HyperCube	Fully Connected	Network of Workstations
Gradient Model	0.72	0.74	0.76	0.54
Sender Initiated Diffusion	0.65	0.65	0.68	0.52
Central Job Dispatcher	0.65	0.66	0.65	0.45
Rate of Change Model	0.74	0.75	0.81	0.60



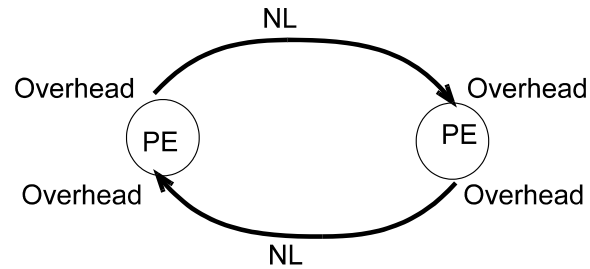
RoC-LB: Results

Symmetric distribution with Arrivals

	Mesh	HyperCube	Fully Connected	Network of Workstations
Gradient Model	0.70	0.77	0.81	0.54
Sender Initiated Diffusion	0.73	0.76	0.81	0.58
Central Job Dispatcher	0.52	0.53	0.53	0.42
Rate of Change Model	0.82	0.82	0.88	0.62



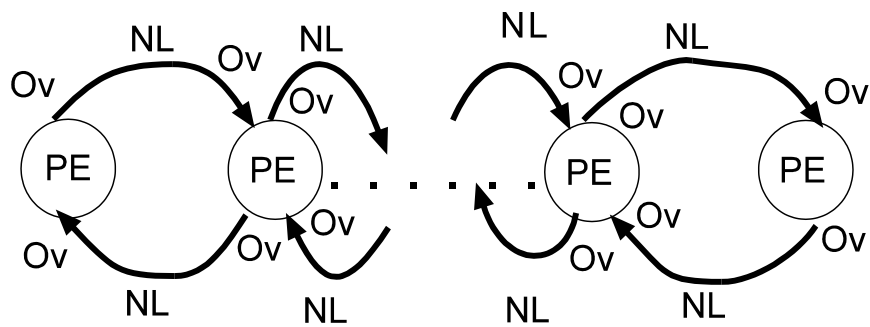
Influence of ND in RoC-LB



$$ND = 4 \times \text{Overhead} + 2 \times \text{Network Latency}$$



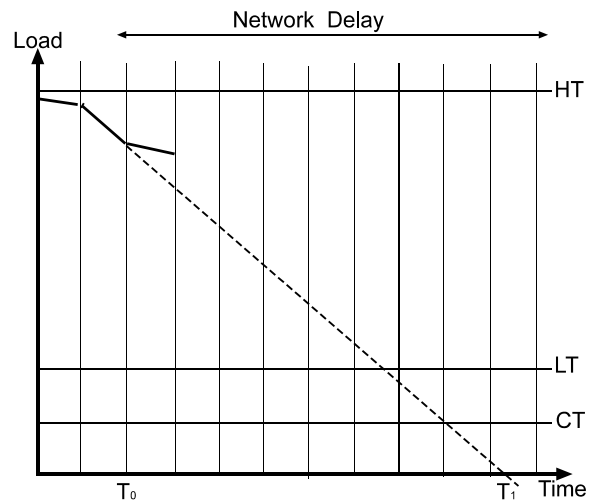
Influence of ND in RoC-LB



$$ND = \text{Number Hops} \times (2 \times NL + 4 \times Ov)$$



Influence of ND in RoC-LB



Very sensitive to even small changes in the load



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RoC-LB: Table information accuracy

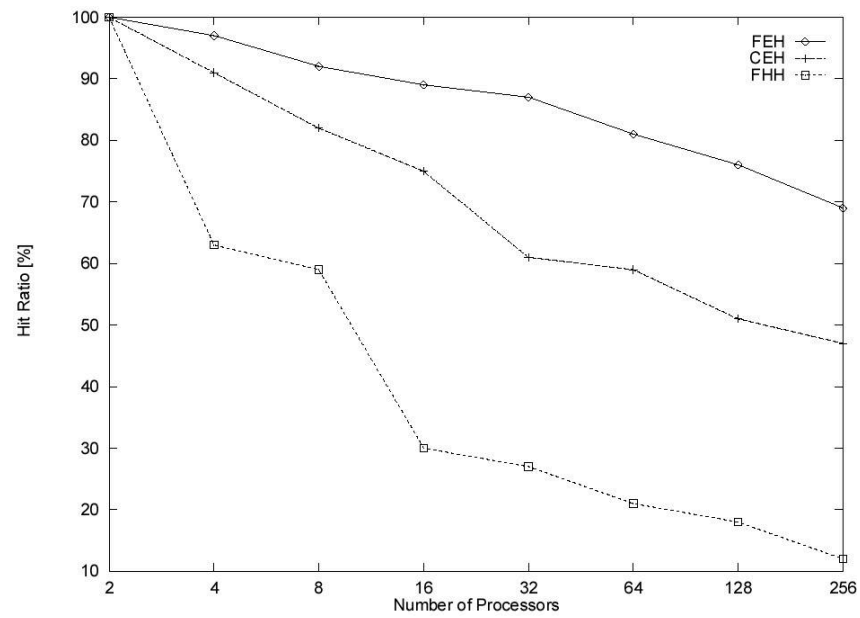
- ▶ **First Hop Hit**
 - ▶ The first entry in the table is accurate in the first hop
- ▶ **First Entry Hit**
 - ▶ The first entry in the table leads to a load transfer
- ▶ **Complete Entry Hit**
 - ▶ The load request is completely satisfied



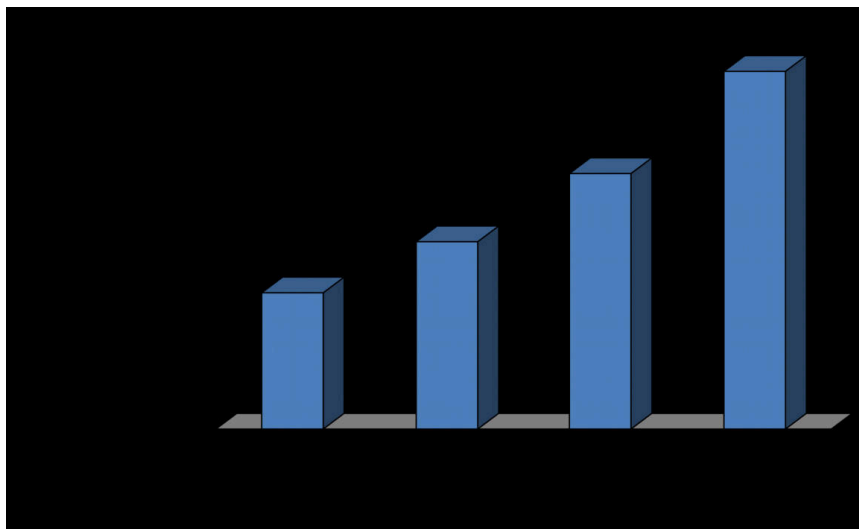
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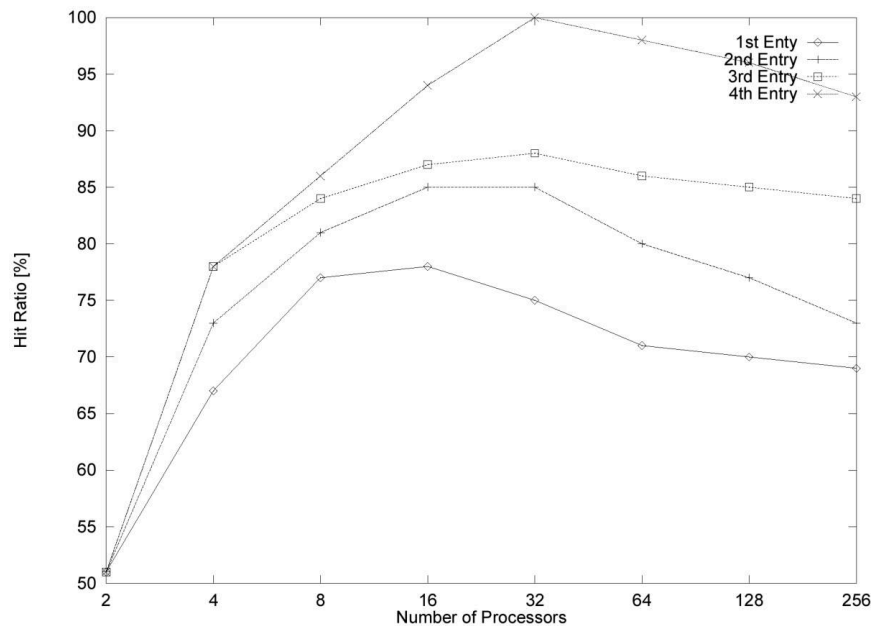
RoC-LB: Results (experiment 1)



RoC-LB: Results (experiment 1)



RoC-LB: Results (experiment 2)



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RoC-LB: Improvements

- ▶ How to increase the probability of a complete hit in shorter time?
 - ▶ **Alternative 1:** Every PE sends a load request to all entries in the source table.
 - ▶ **Alternative 2:** Only the request initiator PE sends a load request to all entries in the source table. Any intermediate PE forwards the message to just one other PE.
 - ▶ **Alternative 3:** The initiator PE multicasts a request to all PEs in its source table.
 - ▶ Each request will contain a different number of loads units or
 - ▶ Each request will be forwarded a different number of times.



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Conjecture

Parallel Scheduling > *Local Scheduling* + *Load Balancing*

*“For every Parallel Schedule, there exist an equally effective
Local Scheduling + Load Balancing”*

